Generowanie danych

Uwaga prez pomyłke nie dałem set seed!

```
n<-300
p<-20
X<-matrix(rnorm(n*p,2,5),ncol=p,nrow=n)
X[,1]<-rep(1,n)
beta<-c(rnorm(10,0,1/4),numeric(10))
pi.p<-1/(1+exp(-X%*%beta))
colnames(X)<-sapply(1:p,function(i){paste0("X",i)})
dane<-data.frame(X)
dane$Y<-rbinom(n,1,pi.p)
write.csv(dane,"daneG5.csv",row.names=F)
```

```
id.0<-which(dane$Y==0)
id.1<-which(dane$Y==1)

train.0<-sample(id.0,size=floor(0.7*length(id.0)))
train.1<-sample(id.1,size=floor(0.7*length(id.1)))

id.train<-sort(c(train.0,train.1))
id.test<-setdiff(seq_len(n),id.train)

train<-dane[id.train,]
test<-dane[id.test,]
```

Konstrukcja modelu

```
model.full<-glm(Y~.,data=train[,-1],family=binomial)
model.null<-glm(Y~1,data=train[,-1],family=binomial)

anova(model.null,model.full,test='LRT')
```

```
## Analysis of Deviance Table
##
## Model 1: Y ~ 1
## Model 2: Y ~ X2 + X3 + X4 + X5 + X6 + X7 + X8 + X9 + X10 + X11 + X12 +
##     X13 + X14 + X15 + X16 + X17 + X18 + X19 + X20
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1       208    250.598
## 2       189     44.366 19   206.23 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

H0 - wszytskie bety są równe zero, H1 - H0 jest fałszywa p-value jest mniejsze od progu istotności zatem mamy podstawę do odzrzucenia hipotezy H0

```
summary(model.full)
```

```
##
## Call:
## glm(formula = Y ~ ., family = binomial, data = train[, -1])
##
```

```
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.16010    0.95508   0.168 0.866871
## X2           0.21591    0.11048   1.954 0.050665 .
## X3          -0.77878    0.27708  -2.811 0.004944 **
## X4           0.05571    0.11363   0.490 0.623932
## X5           0.15973    0.12569   1.271 0.203783
## X6           1.87860    0.52671   3.567 0.000362 ***
## X7           0.63523    0.20785   3.056 0.002242 **
## X8           1.34133    0.43560   3.079 0.002075 **
## X9           0.65917    0.21120   3.121 0.001801 **
## X10          0.08370    0.14513   0.577 0.564138
## X11         -0.14201    0.09601  -1.479 0.139126
## X12          0.12399    0.09942   1.247 0.212362
## X13         -0.06728    0.12035  -0.559 0.576135
## X14         -0.21157    0.11960  -1.769 0.076896 .
## X15         -0.17022    0.12999  -1.309 0.190381
## X16         -0.06185    0.10372  -0.596 0.551006
## X17          0.09063    0.11579   0.783 0.433801
## X18         -0.22643    0.11025  -2.054 0.040007 *
## X19          0.21418    0.11560   1.853 0.063915 .
## X20          0.04204    0.07862   0.535 0.592869
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 250.598  on 208  degrees of freedom
## Residual deviance:  44.366  on 189  degrees of freedom
## AIC: 84.366
##
## Number of Fisher Scoring iterations: 10
```

wartości estmatorów są w kolumnie Estimate a błady standardowe w Std. Error wartości Testy p-value są w kolumnie Pr(>|z|) dal bet od X11 do X20 p-value wychodzi duze co jest zgodne z oczekiwaniami dla zmiennych objasnijących od X1 do X10 wychodzi to róznie

```
confint(model.full)

##                    2.5 %      97.5 %
## (Intercept) -1.771500064  2.11478441
## X2           0.013878452  0.46089625
## X3          -1.477009610 -0.35059089
## X4          -0.163412719  0.30082297
## X5          -0.065508408  0.44431913
## X6           1.099681109  3.24125385
## X7           0.320380697  1.16144150
## X8           0.712918820  2.46463600
## X9           0.334770143  1.19379444
## X10         -0.197224023  0.39149745
## X11         -0.353492844  0.03459325
## X12         -0.065888319  0.33577962
## X13         -0.331038533  0.16192402
## X14         -0.489134754 -0.00323386
## X15         -0.453955011  0.06986716
## X16         -0.296082206  0.12690669
## X17         -0.131594367  0.34989892
```

```
## X18         -0.489174576 -0.03827871
## X19          0.006836821  0.47257294
## X20         -0.112415636  0.20868603

cbind(beta,coef(model.full))

##                      beta
## (Intercept)  0.152584145  0.16010388
## X2           0.067454791  0.21590690
## X3          -0.193228122 -0.77878396
## X4          -0.090931454  0.05570894
## X5           0.043473229  0.15972864
## X6           0.879477400  1.87860188
## X7           0.195945892  0.63522617
## X8           0.581219848  1.34133352
## X9           0.276548090  0.65917230
## X10          0.006362454  0.08370005
## X11          0.000000000 -0.14200629
## X12          0.000000000  0.12398788
## X13          0.000000000 -0.06728224
## X14          0.000000000 -0.21157477
## X15          0.000000000 -0.17021998
## X16          0.000000000 -0.06184535
## X17          0.000000000  0.09063057
## X18          0.000000000 -0.22642759
## X19          0.000000000  0.21418408
## X20          0.000000000  0.04203744
```

Model dośc dobrze estymuje parametry B znaki się zgadzaja

```
coef(model.full)[2]

##        X2
## 0.2159069

exp(coef(model.full))[2]

##       X2
## 1.240987
```

Wzrost X2 o 1 jednostkę zwiększa szansę zdarzenia przez 23 procent

```
coef(model.full)[5]

##        X5
## 0.1597286

exp(coef(model.full))[5]

##       X5
## 1.173192
```

Wzrost X5 o 1 jednostkę zmniejsza szansę zdarzenia o 32 procent

```
pred.test<-predict(model.full,
                   newdata=test,type='response')

wskaźniki<-function(poziom=0.5){
```

```
  pred.class<-ifelse(pred.test>poziom,1,0)
  cm<-table(pred=pred.class,real=test$Y)
  sen<-cm[2,2]/sum(cm[,2])
  spec<-cm[1,1]/sum(cm[,1])
  fpr<-cm[2,1]/sum(cm[,1])
  fnr<-cm[1,2]/sum(cm[,2])
  acc<-sum(diag(cm))/sum(cm)
  miary<-c(sen,spec,fpr,fnr,acc)
  names(miary)<-c("sen","spec","fpr","fnr","acc")
  return(list(cm=cm,miary=miary))
}

M.1<-wskaźniki(0.5)
M.2<-wskaźniki(0.7)

cbind(M.1$miary,M.2$miary)
```

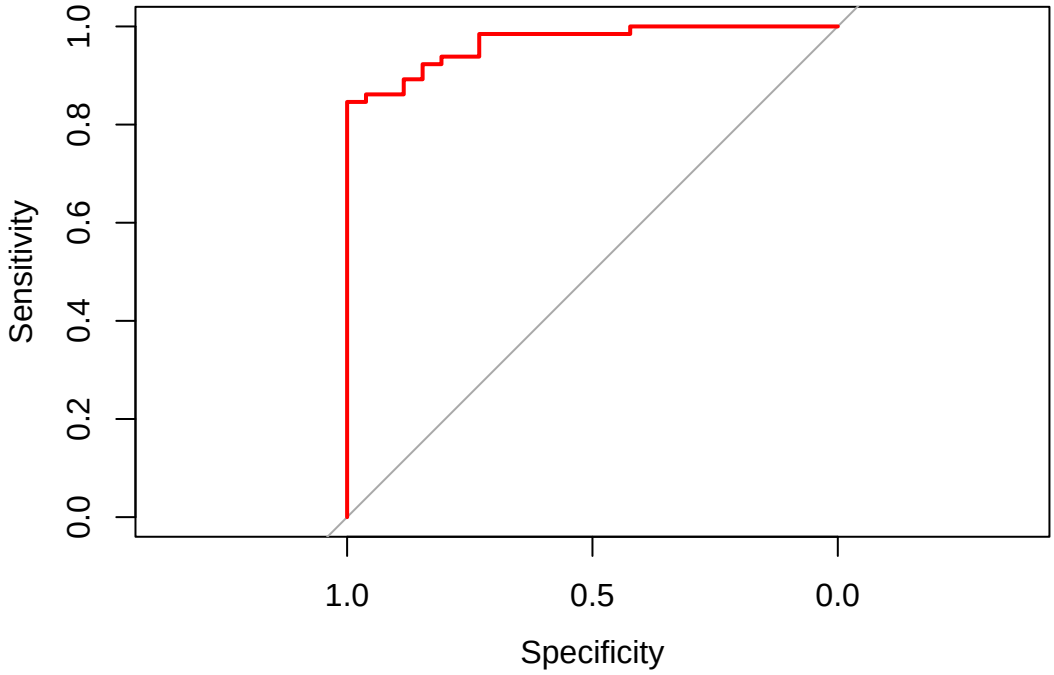
```
##           [,1]      [,2]
## sen  0.8769231 0.8615385
## spec 0.8846154 0.8846154
## fpr  0.1153846 0.1153846
## fnr  0.1230769 0.1384615
## acc  0.8791209 0.8681319
```

Według kryterium Youden M.1 = 0.779, M.2 = 0.837 zatem według tego kryterium M.2 lepszy

```
library(pROC)
roc.obj<-roc(test$Y,pred.test)
auc(roc.obj)
```

```
## Area under the curve: 0.9669
```

```
plot(roc.obj,col='red')
```



AUC jest na poziomie 0.97 co sprawia ze model dość dobrze odróżnia klasy